EL-2003 – Computer Organization & Assembly Language Lab
Lab # 3

Objectives:

- Physical Memory Address Calculation (Segment:Offset)
- Direct & Indirect Addressing Modes

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your program.
2. Write Program in multiple cells of the same notebook**.**
3. **Must Use Functions and Exception handling for each question, where they can be applied.**
4. Do not copy from any source otherwise you will be penalized with negative marks.
5. Add your source code in word document and attach output screenshots.
6. Upload word file with codes and screenshot.
7. Submit your lab on Google Classroom.

Question 1:

Given Segment = 1000h, Offset = 0050h, calculate the physical address using (Segment × 10h) + Offset. Repeat for Segment = 19F5h, Offset = FFFEh. Verify one of your answers against AFD's CS/DS and IP register values on a running program.

Question 2:

Write an ASM program that stores 25 at memory offset 0300h using direct addressing, loads it into AX, adds 15, and writes the result back to 0300h. Attach code, compile screenshot, and an AFD memory-dump screenshot showing the value before and after.

Question 3:

Write an ASM program that loads address 0400h into BX, stores 42 there using [bx], then reads it back into AX using [bx]. In a short note, explain whether BX holds the address or the value at each step.

Question 4:

Write two short programs that both store 99 at offset 0600h and read it back: one using direct addressing, one using indirect addressing via any register. State which used fewer instructions, and describe one situation where indirect addressing would clearly be the better choice.